

Boundary pin / I/O assignment

Pins sit on outline edges

Pseudocode

- Pin assignment validates boundary I/O
- Pseudocode checks side and offset ranges and requires all four outline sides to appear
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- Golden pins cover left, bottom, right, and top and return valid true
- An empty list fails validity

Algorithm sketch — try these

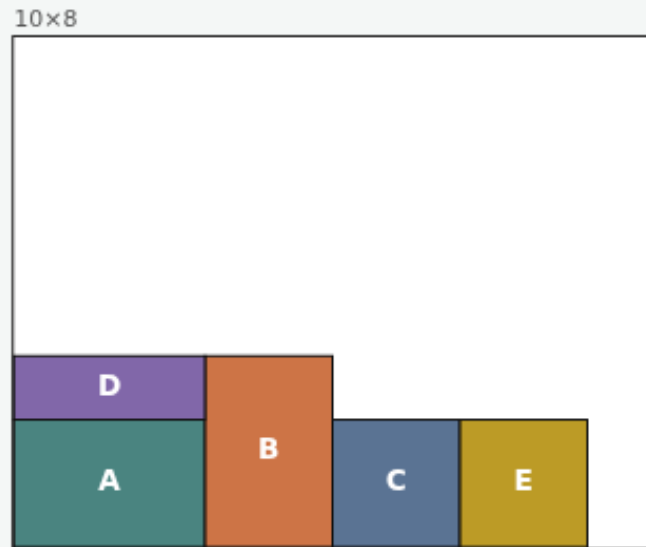
```
INPUT: outline, pin list {side, offset}  
OUTPUT: pinsValid?  
each pin on left|right|top|bottom edge  
offset in range for that side  
require all four sides represented  
GOLDEN 4 pins (one/side) → valid true  
empty list → valid false
```

Packing without pins

PIN ASSIGNMENT

Packing without pins

Step 1 / 5 · no-pins · module04-03-pin-assignment



Explanation

Start from the golden packing with an empty pin list. Pins will sit on outline edges—not inside modules—for I/O literacy.

- Golden pack shown
- pins: none
- Four sides available

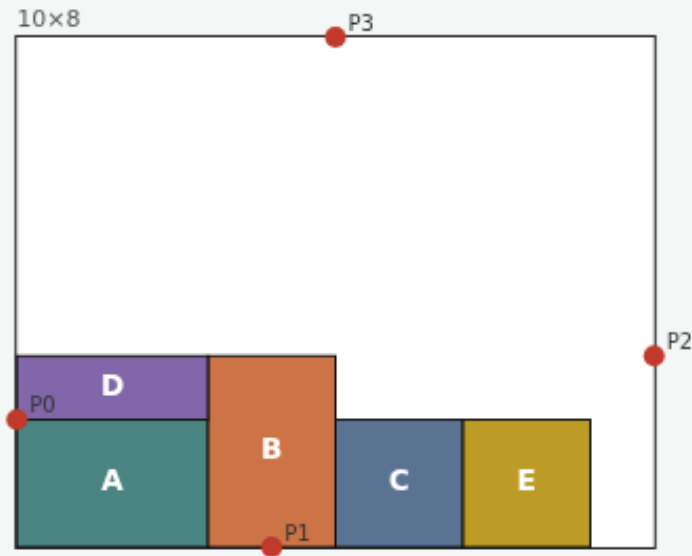
```
pins: 0  
valid: false (needs 4 sides)
```

Assign one pin per side

PIN ASSIGNMENT

Assign one pin per side

Step 2 / 5 · assign · module04-03-pin-assignment



Explanation

Golden pins place P0 on the left, P1 on the bottom, P2 on the right, and P3 on the top. Offsets stay inside each edge length.

- P0 left @2
- P1 bottom @4
- P2 right @3
- P3 top @5

pins: 4
sides: 4

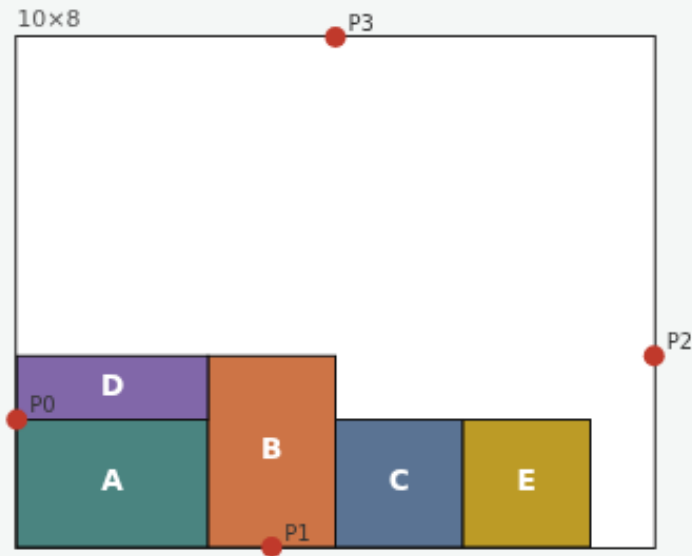
Assign one pin per side

Coverage makes pinsValid true

PIN ASSIGNMENT

Coverage makes pinsValid true

Step 3 / 5 · valid · module04-03-pin-assignment



Explanation

pinsValid requires every side to appear and offsets to lie on the edge. The golden set covers all four sides—valid returns true.

- Four distinct sides
- Offsets in range
- valid: true

```
valid: true  
ids: P0-P3
```

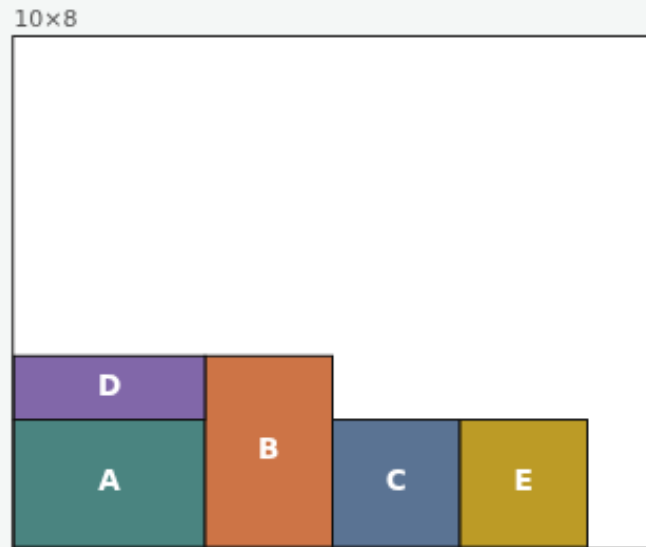
Coverage makes pinsValid true

Empty set is invalid

PIN ASSIGNMENT

Empty set is invalid

Step 4 / 5 · empty-bad · module04-03-pin-assignment



Explanation

Clear the pins and validity fails: you no longer cover four sides. Assignment is a first-class constraint, not decoration.

- Empty → invalid
- Need four sides
- Order/offsets matter later

```
pins: 0  
valid: false
```

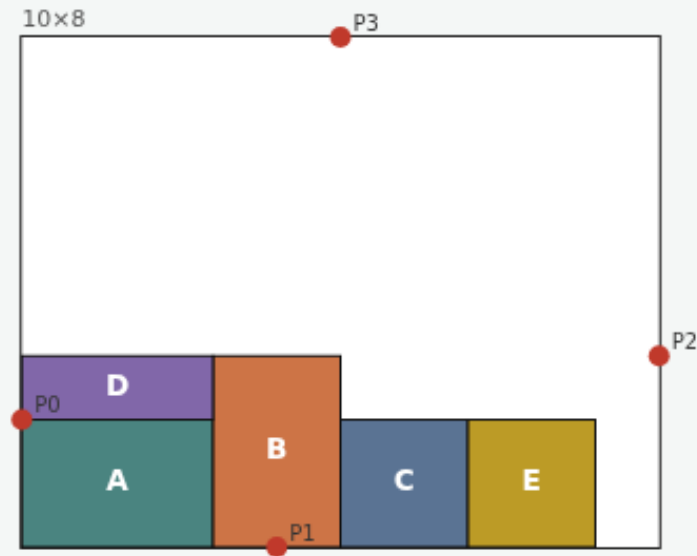
Empty set is invalid

Pins feed place and route

PIN ASSIGNMENT

Pins feed place and route

Step 5 / 5 · takeaway · module04-03-pin-assignment



Explanation

Boundary pins shape terminal propagation and routing demand. After floorplan shapes settle, pin assignment is the bridge to placement.

- One pin per side in the toy
- valid covers left/right/top/bottom
- Next course: learn_placement

starter: P0–P3 on four sides

Pins feed place and route

Browser lab track

- Open pin-assignment
- Assign golden pins, confirm four sides and valid true
- Clear pins and watch validity fail

Implement track

- Implement pinsValid requiring all four sides and in-range offsets
- Assert golden pins pass and the empty list fails

Pitfalls

- Putting pins inside modules

Your turn

- Ship a valid four-side assignment
- Offline compare is next; then the wrap points to learn_placement